

Heredity and Evolution

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Heredity

Gene are responsible for transfer of biological character

Central Dogma of protein synthesis

Mutation

Transcription

DNA → RNA

Translation

RNA → Protein

Translocation

Movement of Ribosome on mRNA 5' → 3'

Evolution

Morphological evidences

Structure of mouth, position of eyes, structure of nostrils & ear pinna.

Anatomical evidences

Human hand, foreleg of ox, flipper of whale & patagium of bat

Vestigial organs

Appendix, ear muscle, wisdom teeth, tail bone, hairs

Palaeontological evidences

1. Carbon dating
2. Willard Libby

Connecting link

1. Peripatus Annelida & Arthropoda
2. Duckbill platypus reptiles & mammals
3. Lung fish Pisces & Amphibia

Embryological evidences

All vertebrates embryo shows similarities at its initial stage of development

Theories of evolution

Darwin's theory

1. Book - 'Origin of Species'
2. Natural selection
3. Survival of the fittest

Lamarckism

1. Use or disuse of organ
2. Theory of inheritance of acquired character
3. e.g. neck of giraffe.

Speciation

Reasons
1. Genetic variation
2. Geographical & Reproductive isolation

Respiration

Aerobic Respiration

Glycolysis

Glucose $\rightarrow$ 2 Pyruvate
+ 2 NADH, + 2ATP + 2H₂O

Kreb Cycle

Acetylco-A $\rightarrow$ 1 ATP +
3 NADH, +1 FADH, +2 CO₂

ETC Reaction

1 NADH, $\rightarrow$ 3 ATP
1 FADH, $\rightarrow$ 2 ATP

Anaerobic Respiration

Lactic Acid Fermentation

Muscle Cell

Alcoholic Fermentation

Yeast Cell

Energy From Different Food Component

Protein

- (1) 1 gm $\rightarrow$ 4 kcal of energy
- (2) Examples
 - (1) Ossein-Bone
 - (2) Skin-melanin, keratin

Lipid

- (1) 1 gm $\rightarrow$ 9 kcal of energy
- (2) Uses
 - (1) Plasma membrane
 - (2) Progesterone,
 - (3) Estrogen

Vitamins

- (1) Fat soluble e.g. A, D, E, K
- (2) Water soluble e.g. B, C

Water

- (1) Body-65-70%
- (2) Cell - 70%
- (3) Blood Plasma - 90%

Cell Division

Mitosis

- (1) From one parent cell two daughter cell get formed
- (2) All daughter cells are diploid (2n)

Phases

Prophase

Metaphase

Anaphase

Telophase

Meiosis

- (1) From one parent cell four daughter cell get formed
- (2) All daughter cells are haploid (n)

Phases

Mitosis

- (1) Prophase I
- (2) Metaphase I
- (3) Anaphase I
- (4) Telophase I

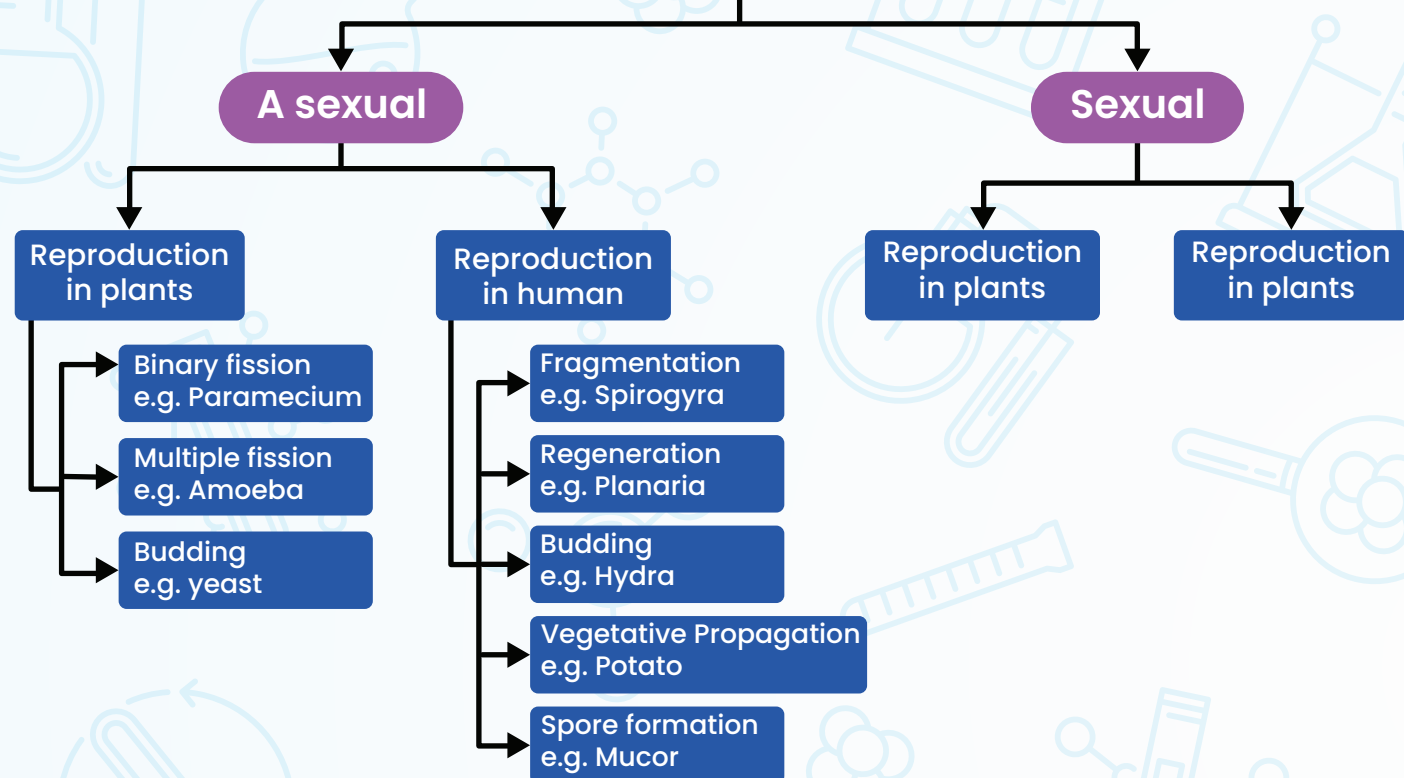
Phases

Meiosis-II

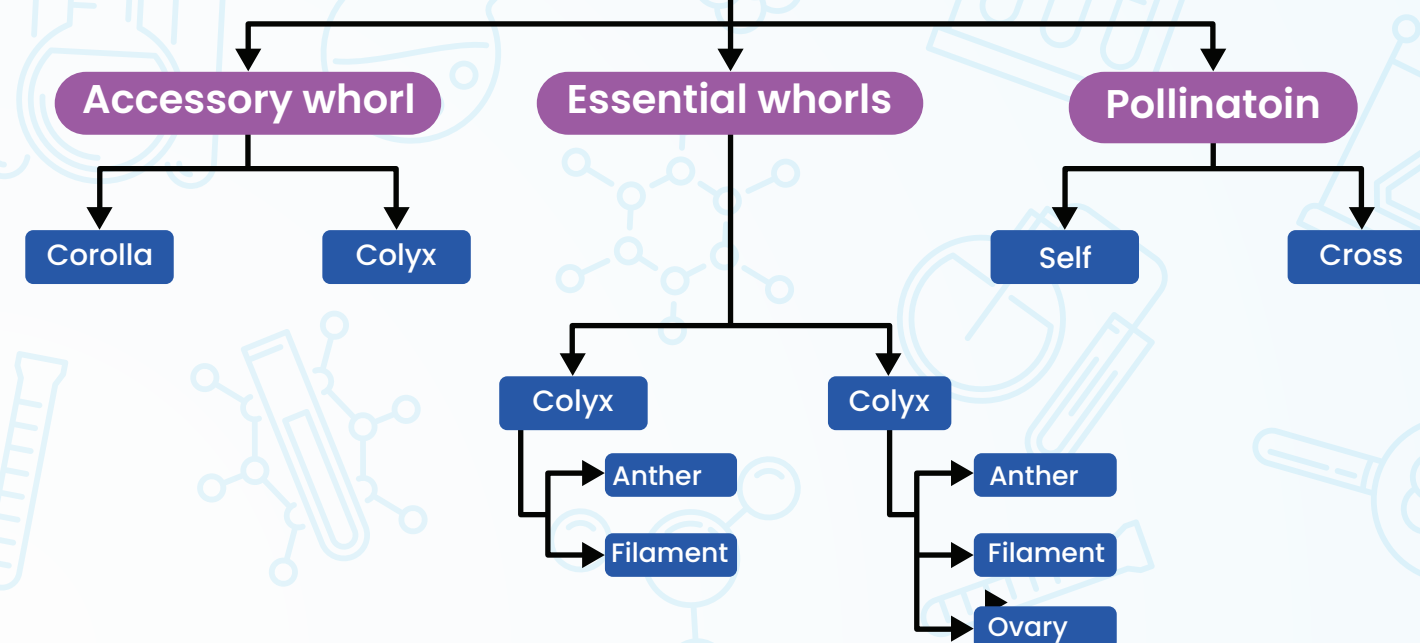
- (1) Prophase II
- (2) Metaphase II
- (3) Anaphase II
- (4) Telophase II

Phases

Reproduction



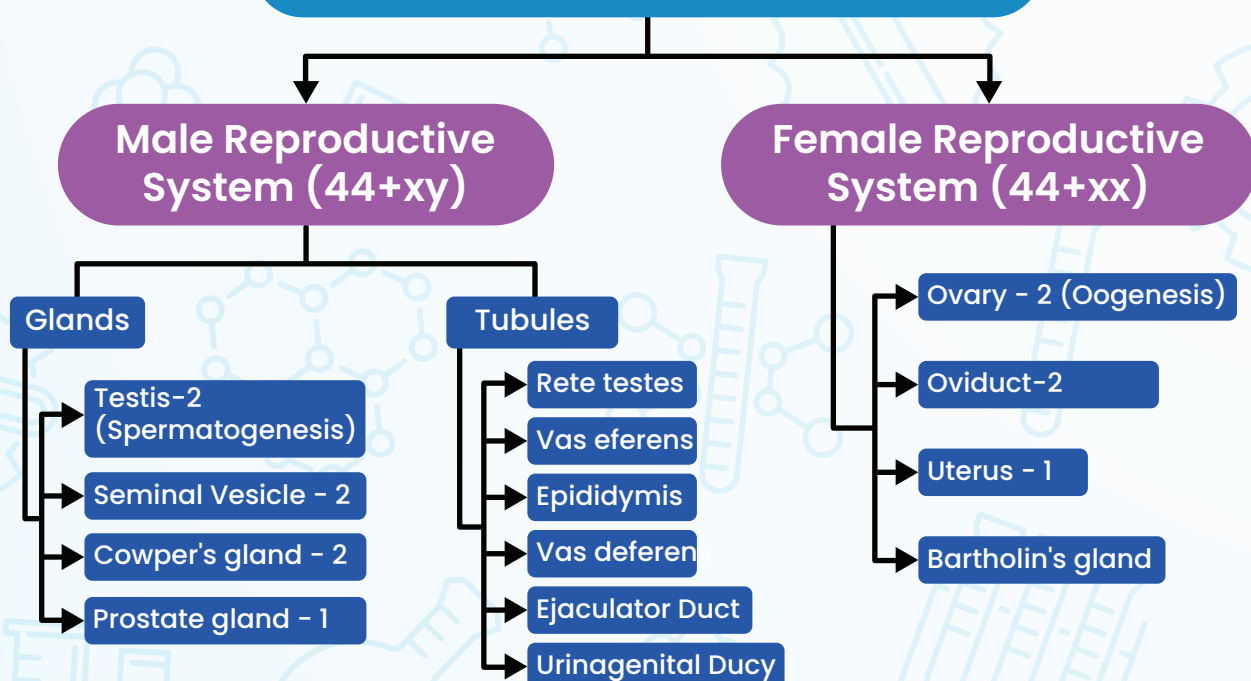
Reproduction in Plants



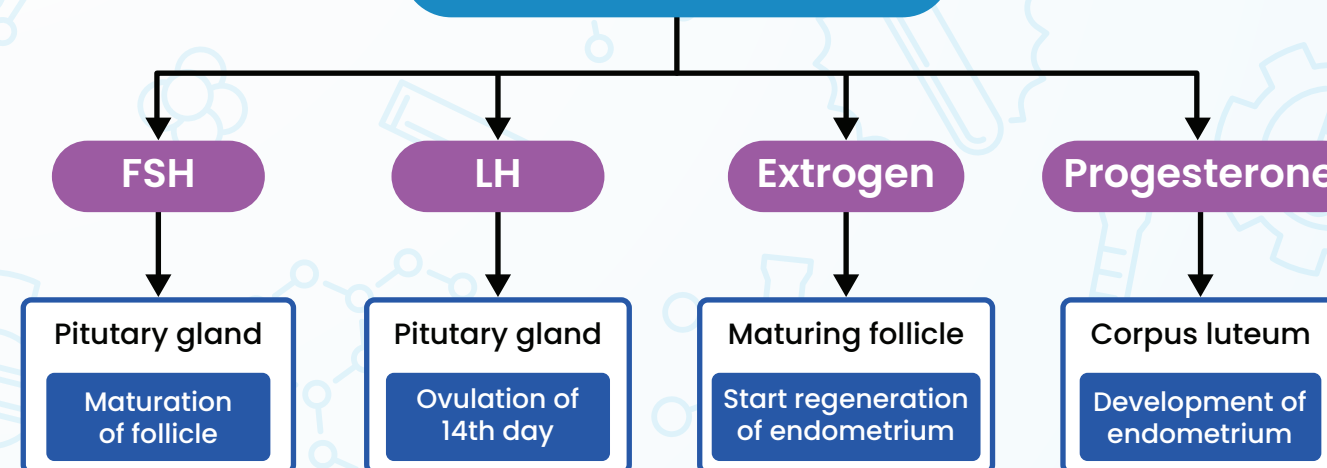
Double fertilization

- 1st male gamete + egg cell = zygote → embryo
- 2nd male gamete + polar nuclei = Endosperm

Reproduction in Humans



Menstrual cycle



Environment

Abiotic

Physical

- Temp
- Humidity
- Sunlight
- Soil
- Climate

Chemical

- Water
- Proteins
- Carbohydrates
- Salt
- Lipids

Evolution

Producers

- Consumers
- Primary
- Secondary
- Tertiary
- Apex
- Climate

Biodiversity

- Genetic
- Species
- Ecosystem

ENVIRONMENTAL PROBLEMS

Abiotic

Pollution

- Water
- Air
- Land

Urbanization

Industrialization

Evolution

Deforestation

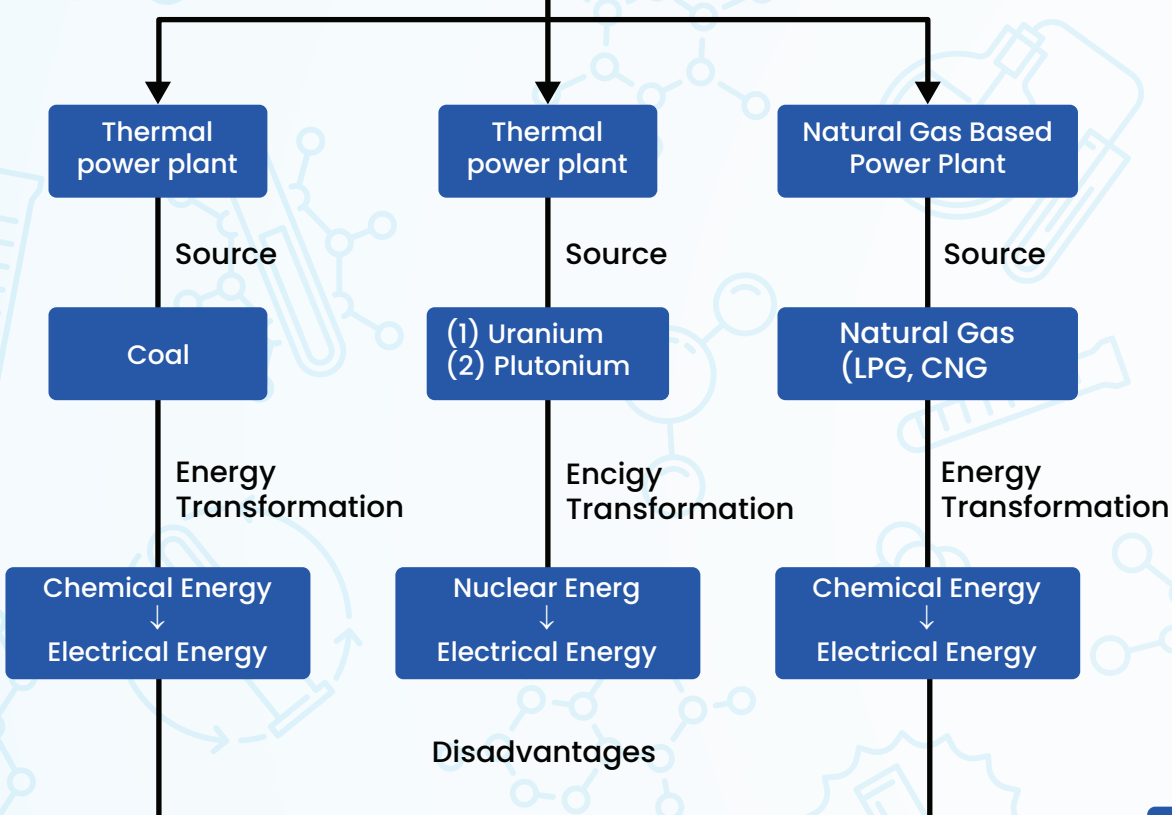
- Climate change
- Global warming

Biodiversity

- Endangered
- Rare
- Vulnerable
- Indeterminate

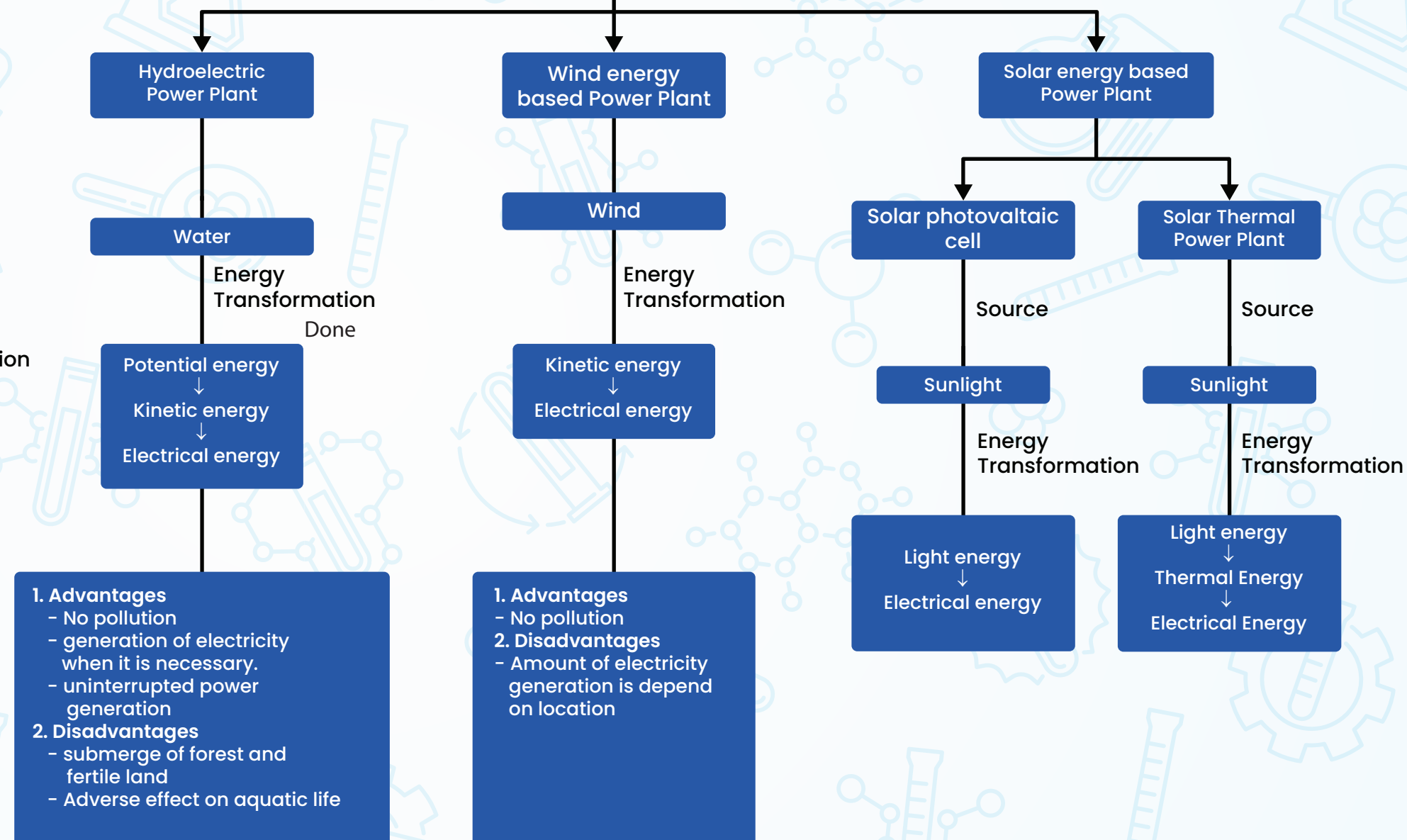
Source of Energy

Non-Renewable Source of energy



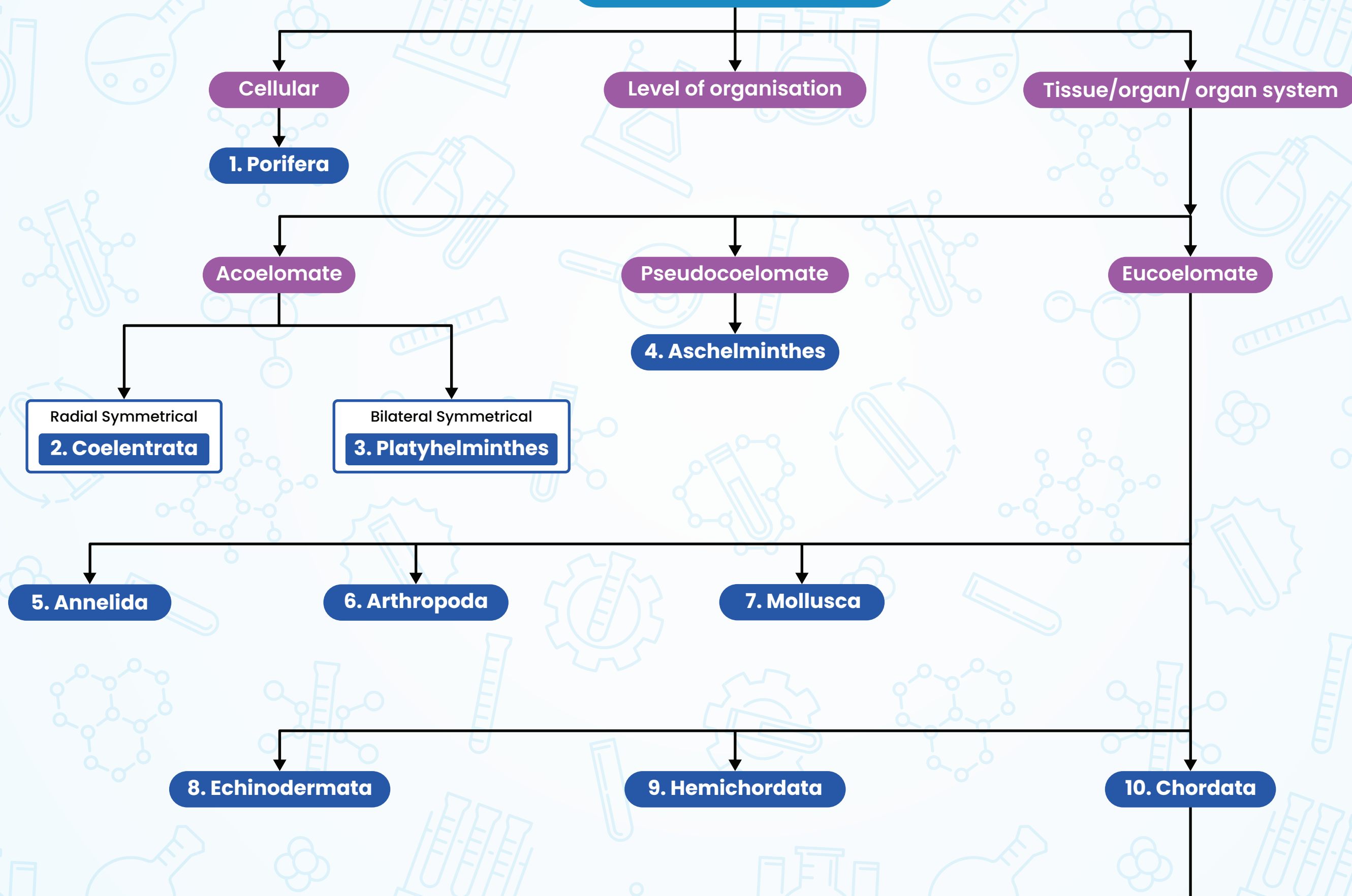
- (1) Air Pollution
- (2) Acid Rain
- (3) Limited reserves of Fossil fuel
- (4) Radioactive Pollution

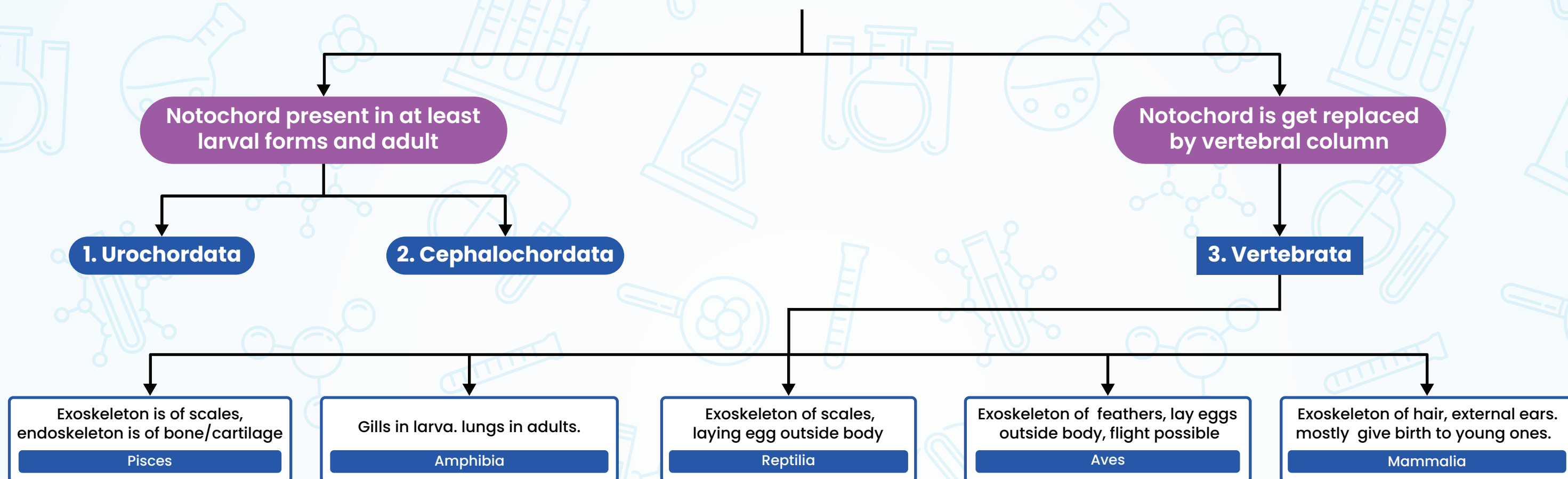
Renewable Source of energy



Animal classification

KINGDOM: ANIMALIA





Applied Microbiology

Proteins

Molecular Biology

Applied Genetics

Study of enzyme related to some prokaryotic and eukaryotic microbes.

Milk

Butter

Yoghurt

Probiotics

Cheese

Fermentation of carbon compounds

Yeast

Saccharomyces cerevisiae

Ehtanol

Microbial degradation

Acetobacter + Glucanobacter

Acetic acid

Bleaching

Potassium ferrocynide

SO₂

Vinegar

Biofuels

Solid

Liquid

Liquid

Coal

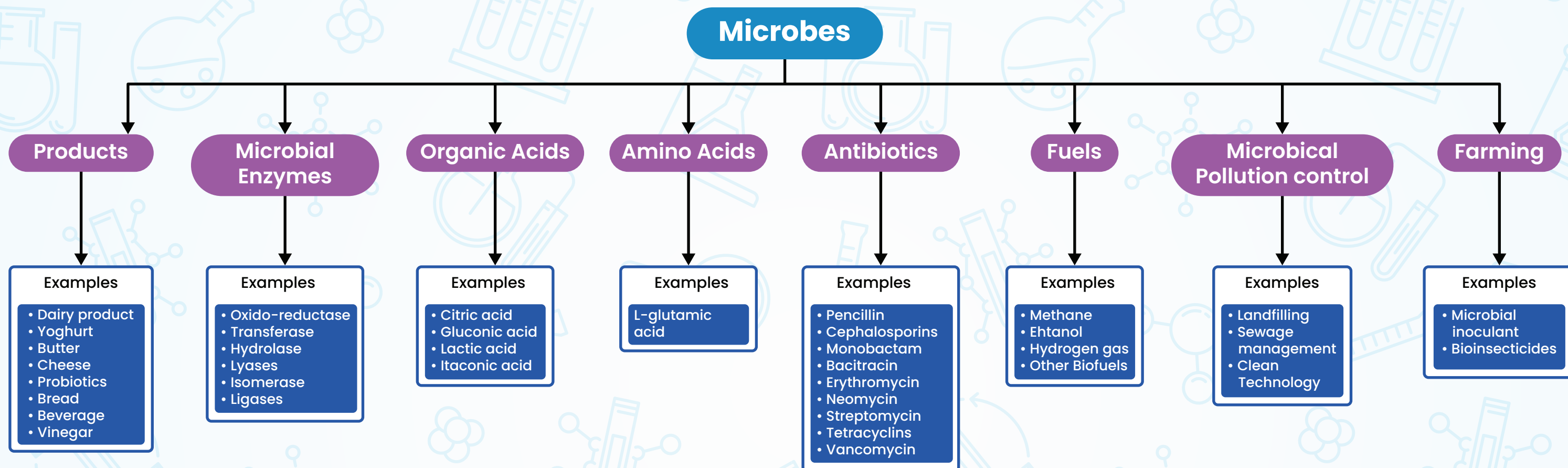
Dung

Alcohol

Vegetable oil

Alcohol

Vegetable oil



Stem Cell

Source

Embryonic Stem cell

Adult stem cell

Uses

Regenerative therapy

Organ Transplantation

Organ Donation

Biotechnology

Commercial Application

Crop Biotechnology

Animal Husbandry

Human Health

Environment & Biotechnology

Food Biotechnology

DNA Fingerprinting

Industrial Products

Hybrid Seed

GM Crops
BT cotton
BT Brinjal
Golden Rice
Herbicide Tolerant Plant

Bio Fertilizers

Vaccines
Edible Vaccine

Treatment

Gene Therapy

Cloning

Reproductive Cloning

Reproductive Cloning

Important Stages in Agricultural Development

Green Revolution

White Revolution

Blue Revolution

Organic Farming

Fruit Processing

Insecticide

Cultivation of Medicinal Plants

Fertilizers

NKM-16

N, P, K

Factors affecting Social health

Mental Stress

Addiction

Chronic diseases

Communication media & excess use of modern Technology

Cyber crime

Stress management

Join laughter Club

Communication With Friends, Peers, teachers, & Parents

Fostering Hobbies

Playing out door games

Regular exercising

Massaging

Yoga

Meditation

Counselling & Psychotherapy

Organizations/schemes

Playing out door games

Work against Tobacco consumption

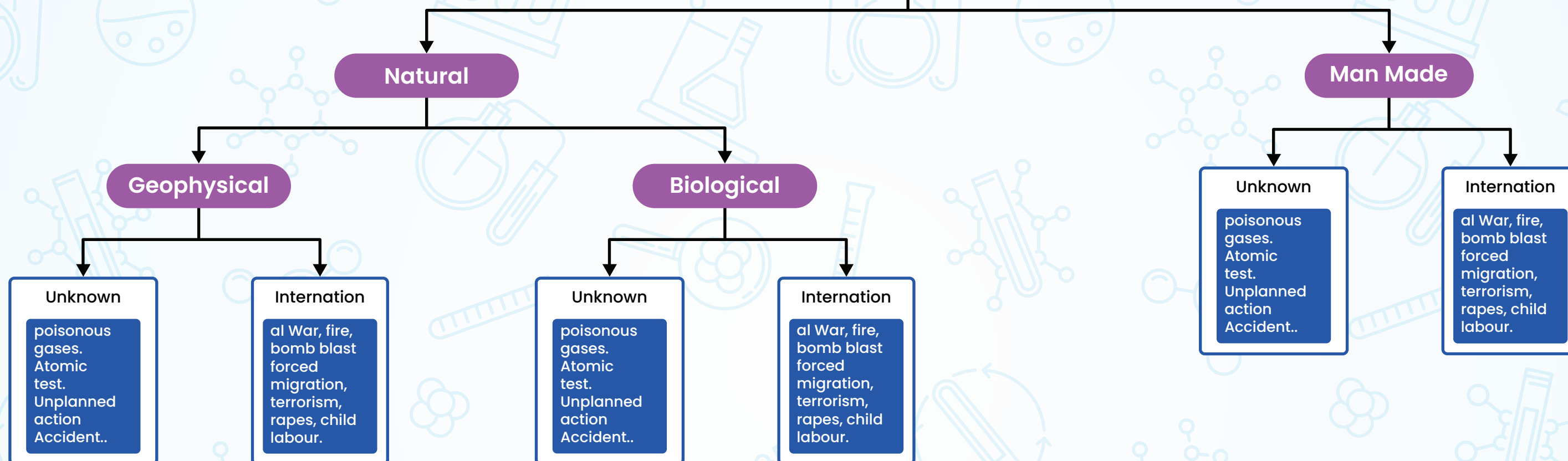
Playing out door games

Empower the Children living in Slum area in the Field of Education, Sports, Arts & Business

Playing out door games

Helpline number For Counselling

Types of Disasters



PHASESE OF DISASTER MANAGEMENT

